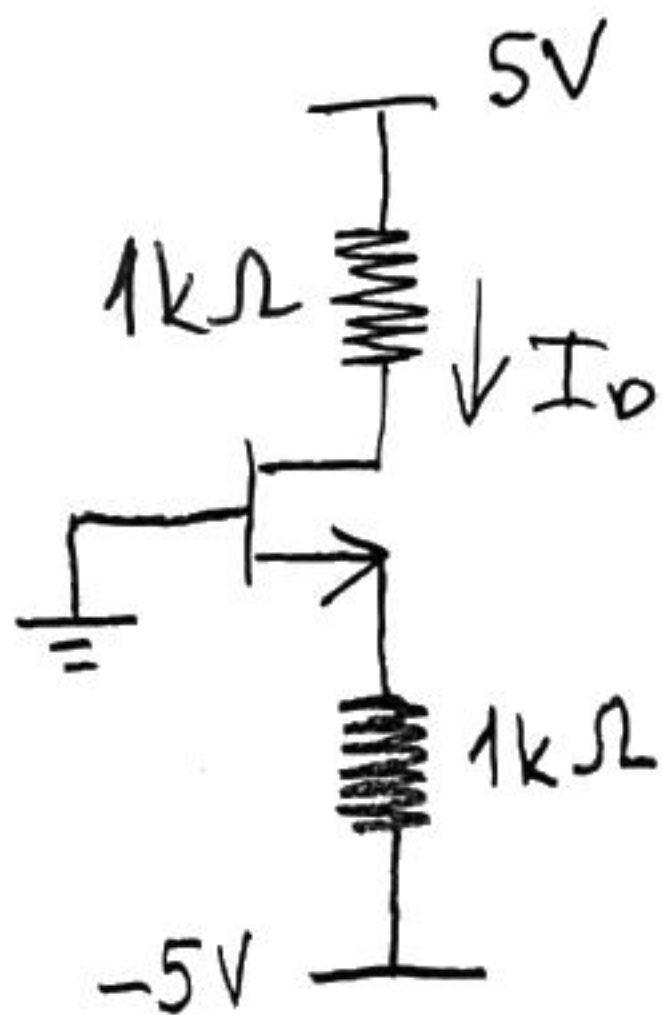


Q1) Find the operating point of the circuit below:

Assumption: the nmos is in saturation. ①



$$V_{th} = 2V$$

$$\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) = K = 0.4 \text{ mA/V}^2$$

S1) nmos in saturation, thus;

$$I_D = K (V_{GS} - V_{th})^2$$

* Gate-Source KVL loop:

$$0 - V_{GS} - 1000 I_D = -5 \rightarrow I_D = (5 - V_{GS}) / 1000$$

* Drain-Source KVL loop:

$$5 - 1000 I_D - V_{DS} - 1000 I_D = -5$$

rearranging both equations;

$$V_{DS} = 10 - 2000 I_D \quad \text{and} \quad I_D = (5 - V_{GS}) / 1000 = K (V_{GS} - V_{th})^2$$

$$0.4 \text{ mA/V}^2 (V_{GS} - 2)^2 = \frac{5 - V_{GS}}{1000} \Rightarrow \boxed{\begin{array}{l} V_{GS1} = 3.76V \\ V_{GS2} = -2.26V \end{array}}$$

nmos in saturation according to assumption:

(2)

$$V_{GS} > V_{th} = 2V \rightarrow \boxed{V_{GS} = 3.76V}$$

Insert V_{GS} value into Gate-Source KVL equation:

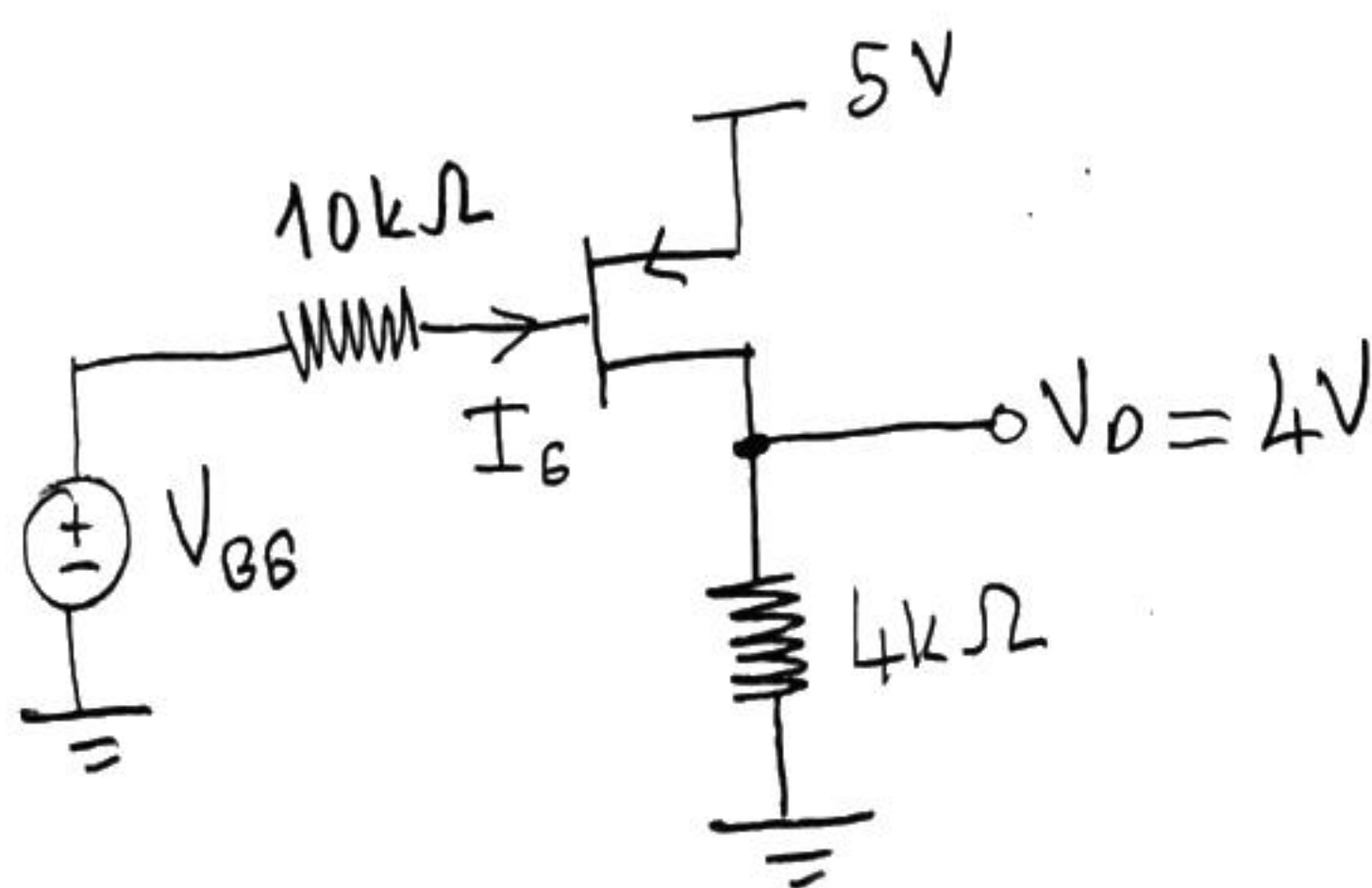
$$I_D = (5 - 3.76)/1000 \rightarrow \boxed{I_D = 1.24 \text{ mA}}$$

Using Drain-Source KVL loop:

$$V_{DS} = 10 - 2000 \cdot 1.24 \cdot 10^{-3} \rightarrow \boxed{V_{DS} = 7.52V}$$

$$\boxed{V_{DS} = 7.52V > V_{GS} - V_{th} = 1.76V}$$

Q2)



$$K = 0.2 \text{ mA/V}^2$$

$$V_{th} = -2V$$

Find V_{GG} !

Assumption = pmos in saturation

$$s2) I_D = K (V_{GS} - V_{th})^2$$

$$I_G = 0 \text{ mA}$$

$$I_D = \frac{V_D - 0}{4000} = \frac{4}{4000} = 1 \text{ mA}$$

$$V_{DS} = V_D - V_S = 4 - 5 = -1V$$

$$V_{GS} = V_G - V_S = V_{GG} - 5$$

$$I_D = K (V_{GS} - V_{th})^2$$

(3)

$$1 \text{ mA} = 0.2 \cdot 10^{-3} (V_{GS} - (-2))^2$$

$$V_{GS_1} = 0.24 \text{ V} \quad V_{GS_2} = -4.23 \text{ V}$$

Assumption: pmos in saturation $\rightarrow V_{GS} < V_{th}$

$$\boxed{V_{GS} = -4.23 \text{ V}}$$

$$V_{GS} = V_G - V_S = V_{GG} - 5 \text{ V} \rightarrow -4.23 = V_{GG} - 5 \text{ V}$$

$$V_{GG} = 0.77 \text{ V}$$

$$V_{GS} = -4.23 \text{ V} < V_{th} = -2 \text{ V}$$

$$V_{DS} = -1 \text{ V} < V_{GS} - V_{th} = -2.23 \text{ V}$$

our assumption was incorrect!

← This is correct only if our original assumption was correct. Thus, we must check to see if our inequalities are valid!

New assumption: pmos in triode region:

$$I_D = K [2 (V_{GS} - V_{th}) V_{DS} - V_{DS}^2]$$

$$I_G = 0 \text{ mA}$$

$$V_S = 5 \text{ V}$$

$$I_D = \frac{V_D - 0}{4000} = 1 \text{ mA}$$

$$V_{DS} = -1 \text{ V}$$

$$V_{GS} = V_G - V_S = V_{GG} - 5 \text{ V}$$

$$1 \text{ mA} = 0.2 \cdot 10^{-3} [2 (V_{GS} - (-2)) (-1) - (-1)^2]$$

$$\boxed{V_{GS} = -5 \text{ V}}$$

Therefore

(4)

$$V_{GG} = V_{GS} + 5 = -5 + 5 = 0V$$

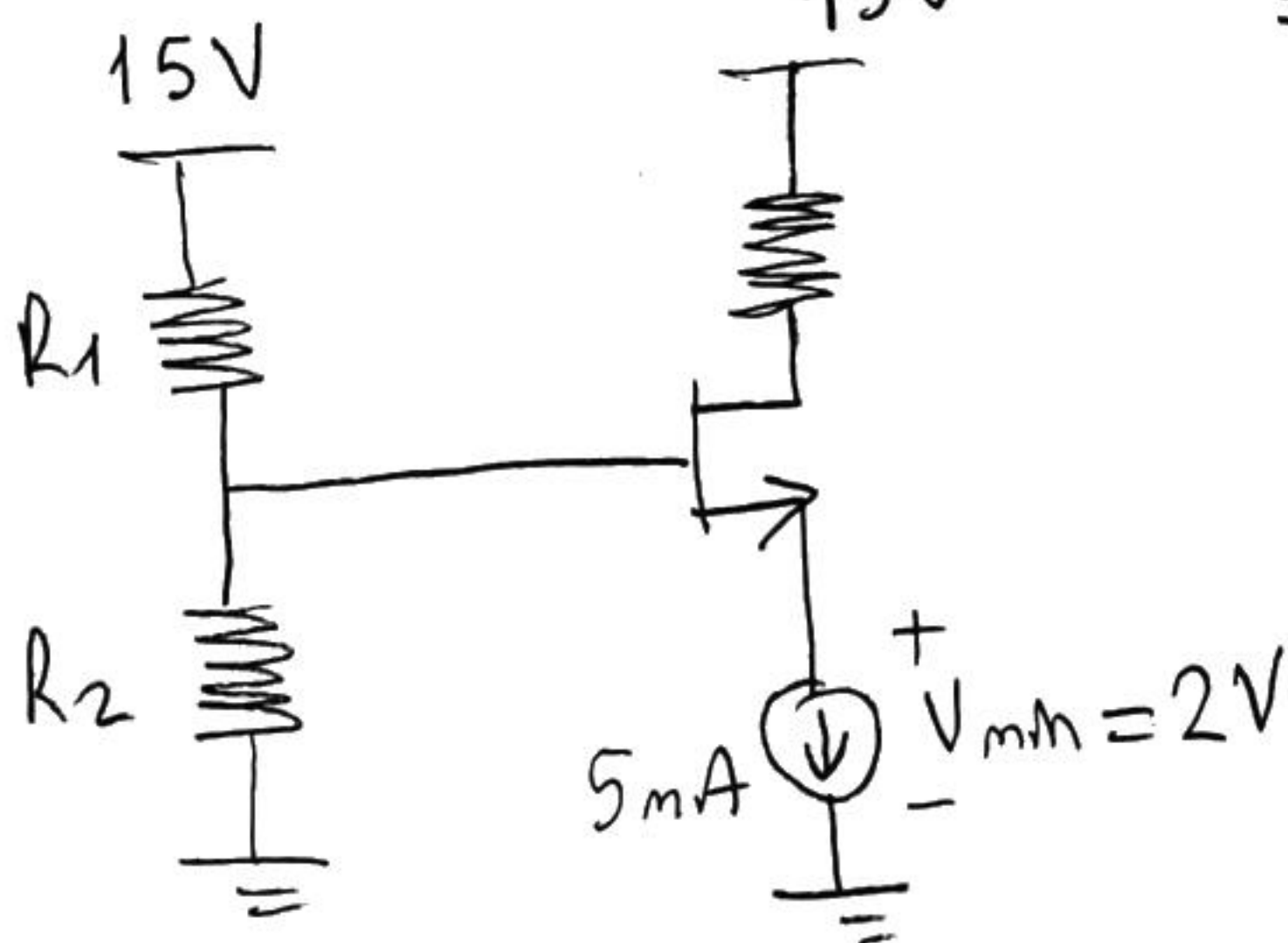
$$V_{GS} = -5V < V_{th} = -2V \Rightarrow \text{Our channel is induced! } \checkmark$$

$$V_{DS} = -1V > V_{GS} - V_{th} = -3V \Rightarrow \text{no pinch-off } \checkmark$$

* Triode assumption is correct!

$$* V_{GG} = 0V$$

Q3) Determine proper resistor values to bias the nmos. $V_{th} = 1V$.



* V_{DS} voltage for the current sink to operate properly!

$$V_{GS} = \sqrt{\frac{I}{K}} + V_{th} = \sqrt{\frac{5}{0.2}} + 1$$

$$V_{GS} = 6V$$

$$V_G = V_{GS} + V_S = 6 + 2 = 8V$$

Select R_1 and R_2 such that $V_G = 8 = V_{DD} \frac{R_2}{R_1 + R_2}$

$$R_1 = 140k \quad R_2 = 160k\Omega$$

→ one possible solution!

$$V_D = 15 - I_D \cdot R_D$$

(5)

$$V_{DS} = V_D - V_S \gg V_{GS} - V_{th} = 6 - 1 = 5V$$

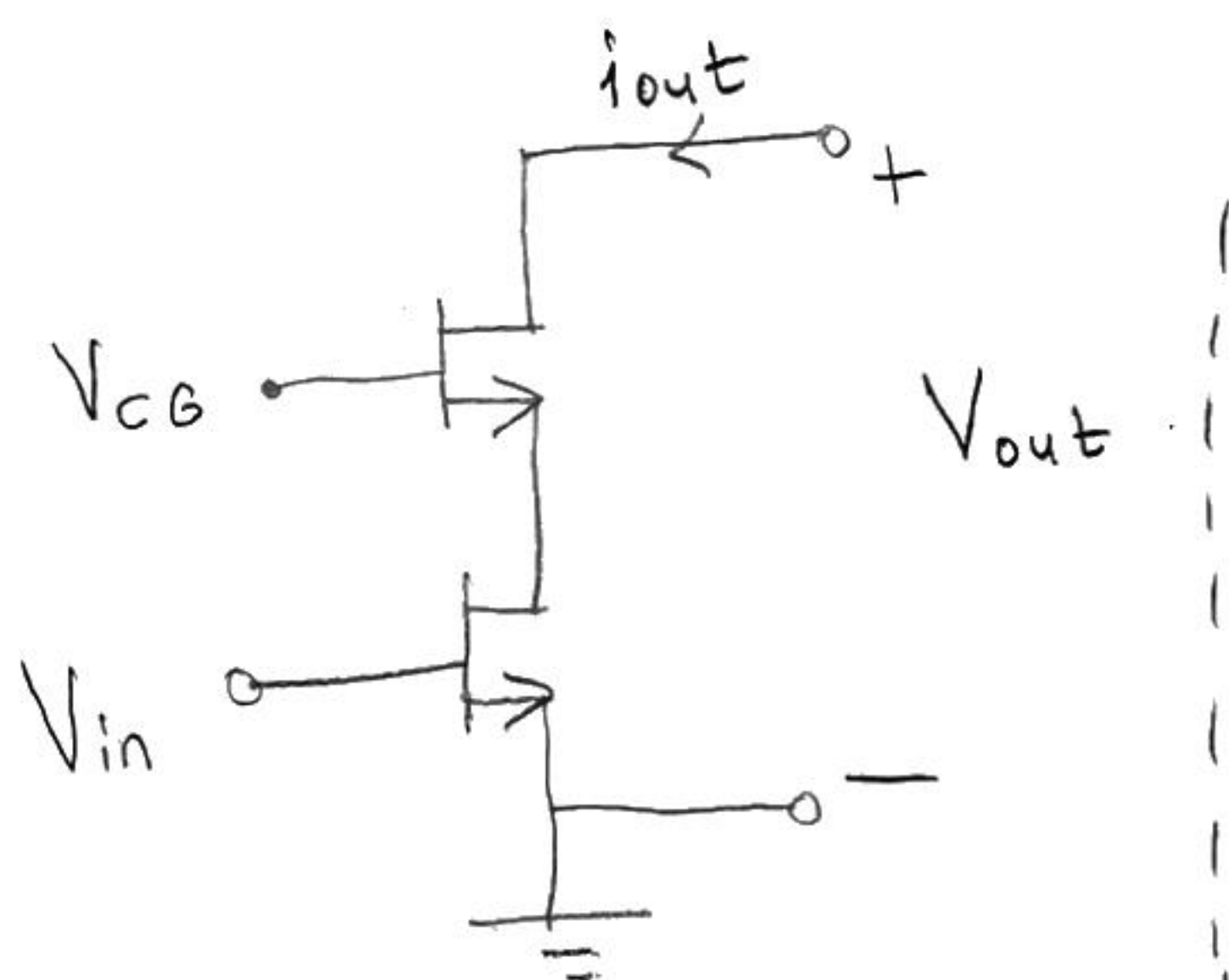
$$V_D \gg V_S + 5 = 2 + 5 = 7V$$

$$15 - 0.005 \cdot R_D \gg 7 \Rightarrow R_D \leq \frac{8}{0.005} = 1600 \Omega$$

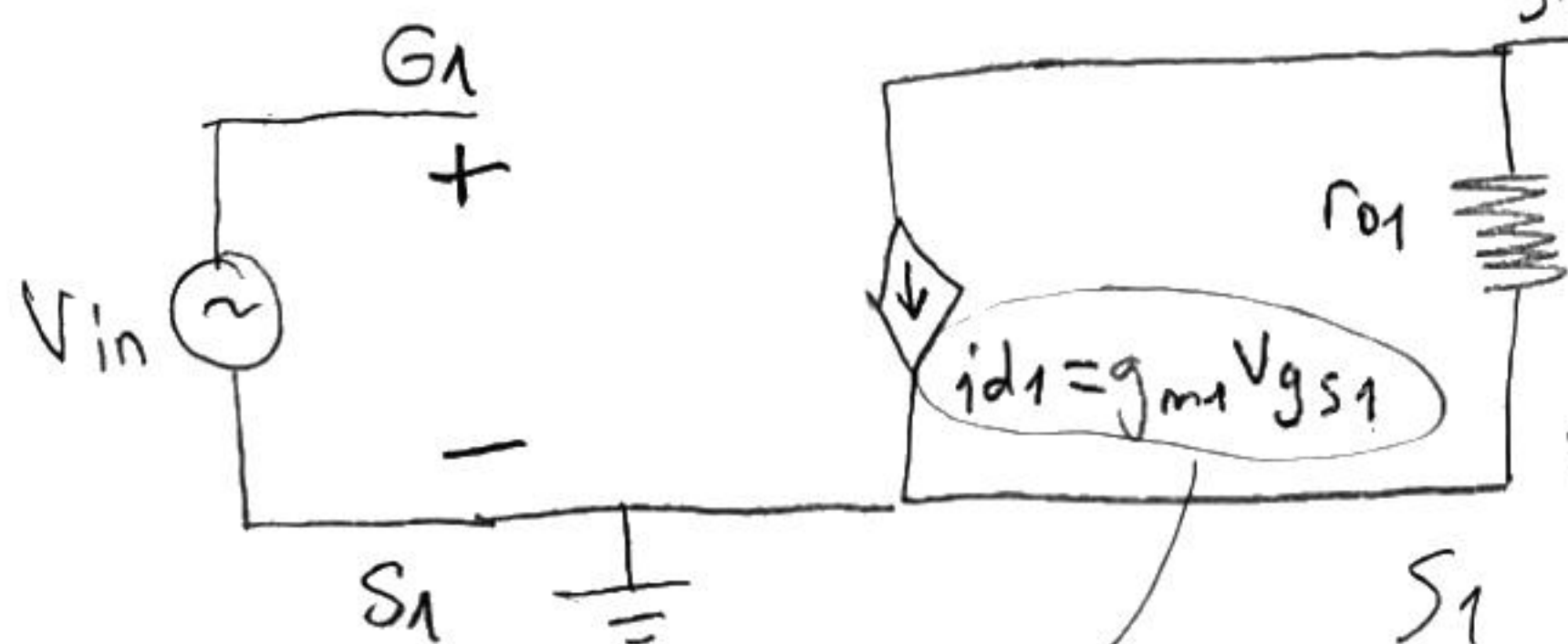
$$\text{e.g. } R_D = 1.5 k\Omega$$

Q 4) Calculate the output resistance if all transistors are in saturation. The small-signal equivalent circuit is also given---

$$r_{out} = \frac{V_{out}}{i_{out}} \bigg|_{V_{in}=0}$$



for small-signal analysis
 $V_{g2} = 0V!$



$$V_{in} = V_{gs1} = 0V$$

$$\Rightarrow V_{gs2} + i_{out} r_{O1} = 0$$

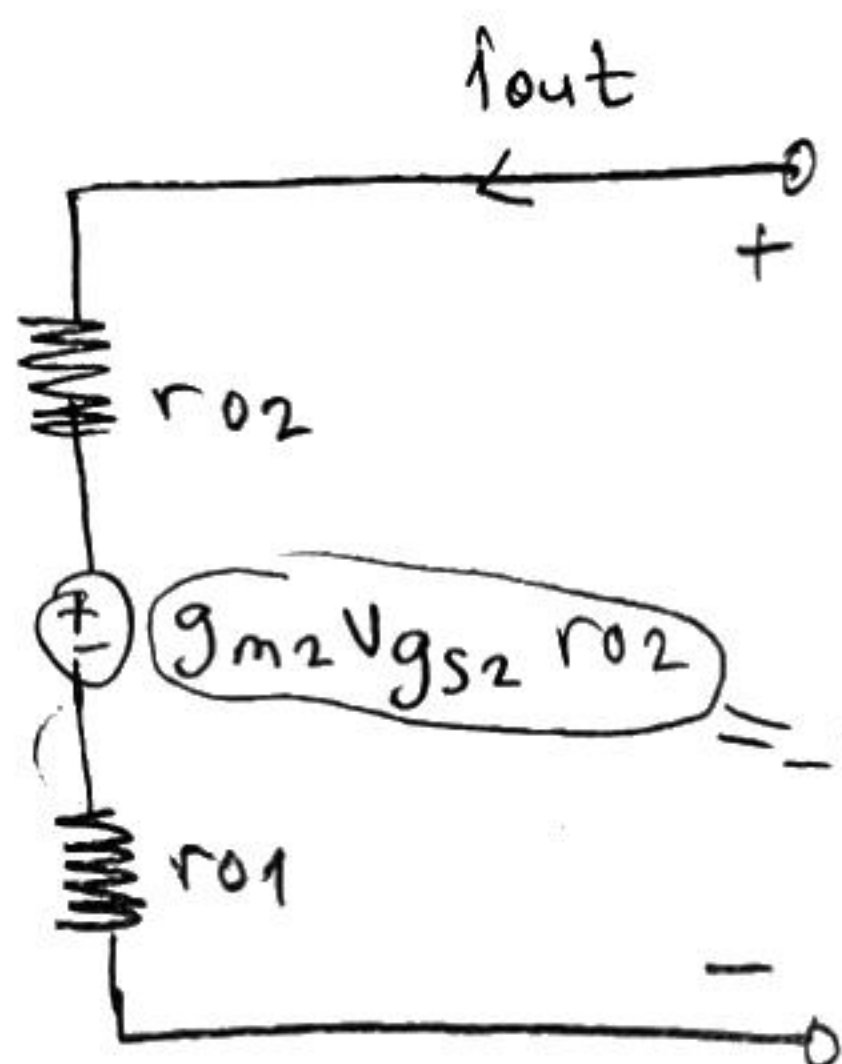
$$* V_{gs2} = -i_{out} r_{O1}$$

Small-signal equivalent circuit of cascode circuit

S4)

6

$$V_{GS1} = 0V$$



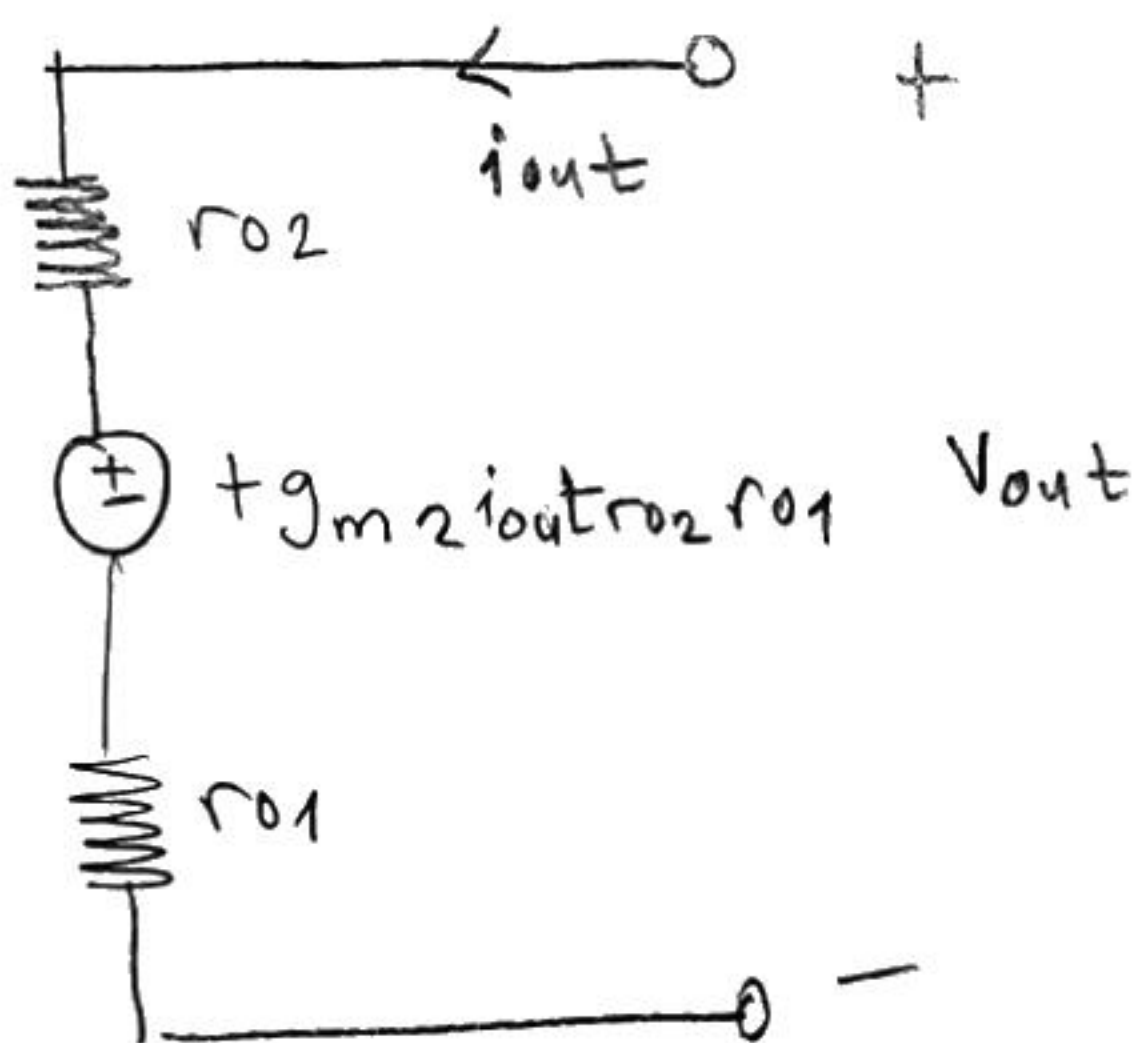
$$r_{out} = \frac{V_{out}}{i_{out}} = \frac{V_{out}}{i_{out}}$$

$$r_{out} = r_{o1} + g_m V_{GS2} r_{o2} + r_{o2}$$

$V_{G2} = 0V$ (see previous page!)

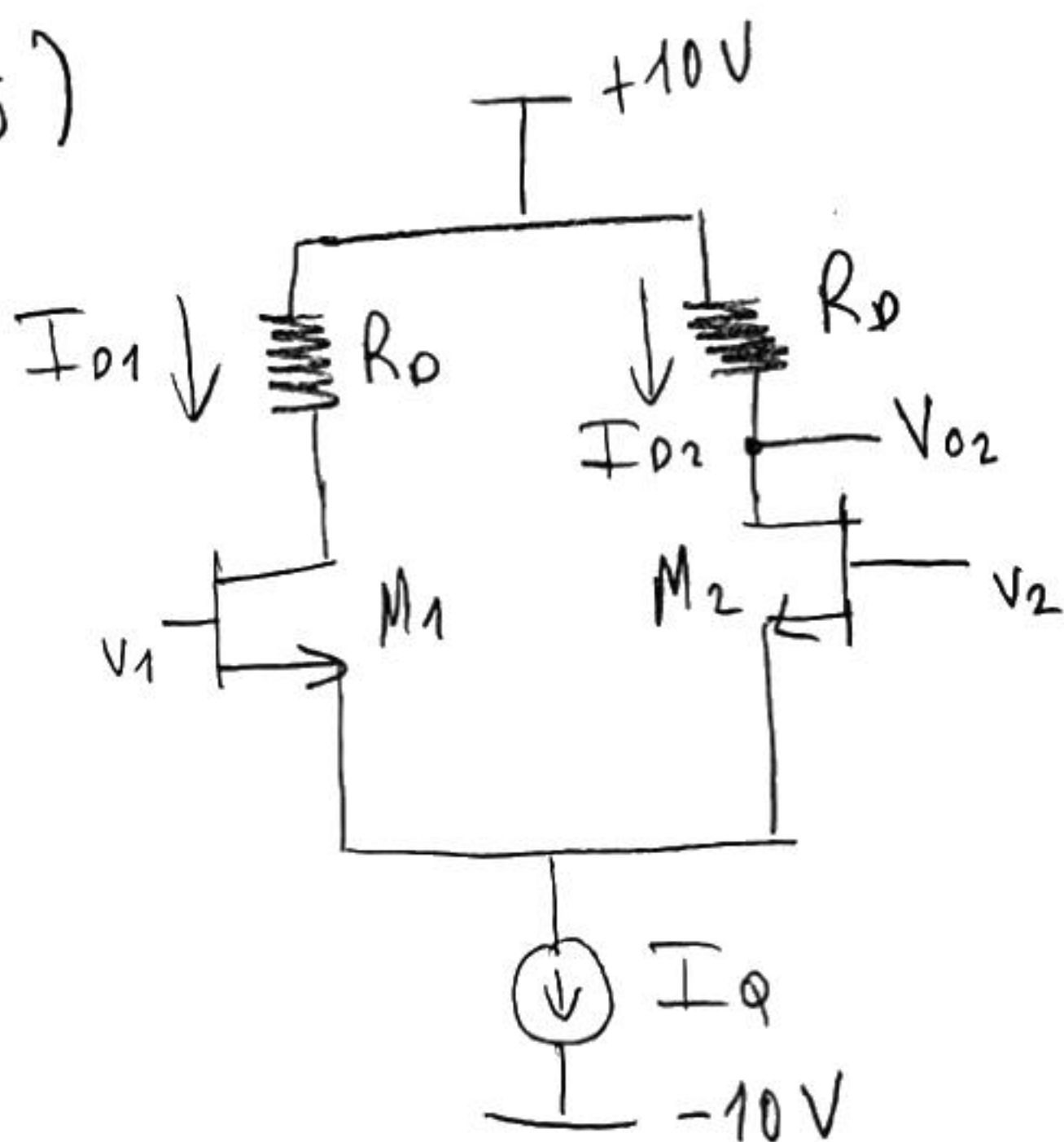
$$g_m V_{GS2} r_{o2} = -g_m V_{S2} r_{o2}$$

$$= -g_m (-i_{out} r_{o1}) r_{o2}$$



$$r_{out} = r_{o1} + r_{o2} + g_m r_{o1} r_{o2}$$

Q5)



Transistor parameters:
 $V_{th} = 0.5V$

$$K = 160 \mu A/V^2$$

a) I_Q , R_D and V_{GS} for
 $I_{D1} = I_{D2} = 80 \mu A$ and
 $V_{O2} = 2V$ if $v_1 = v_2 = 0V$

b) What is the maximum common-mode input voltage?

a) $I_a = I_{D1} + I_{D2}$

$$I_a = 160 \mu A$$

(7)

$$I_D = K (V_{GS} - V_{th})^2 \rightarrow V_{GS} = \sqrt{\frac{I_D}{K}} + V_{th}$$

$$10 = R_D \cdot 80 \mu + 2$$

$$R_D = 100 k\Omega$$

$$= \sqrt{\frac{80 \mu}{160 \mu}} + 0.5$$

$$V_{GS} = 1.207 V$$

b) $V_{CM, max}$

if M_1 and M_2 in saturation, then

$$V_{DS} \geq V_{GS} - V_{th} \rightarrow \underbrace{V_D - V_S}_{V_{O2}} \geq \underbrace{V_G - V_S}_{V_{CM}} - V_{th}$$

$$V_{O2} \geq V_{CM} - V_{th}$$

$$V_{CM} \leq \underbrace{V_{O2} + V_{th}}_{2 + 0.5 V} = 2.5 V$$